Universal property of the cartesian product: the map from functions into the product to pairs of functions is an equivalence

The Univalent Foundations Program

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Editorial note. The counted statement is the author theorem labelled thm:prod-ump (source lines 2372-2375); the proof span 2376-2390 is the author's proof environment, retained line for line.

Editorial note. Function extensionality is used as a background principle and is declared in the standard prerequisites; the book itself introduces the principle in its later material.

Editorial note. Both retained contexts carry the citations used by the proof, so all references resolve inside the composed document.

Editorial scope. One original result of the HoTT book chapter cha:basics, counted once: the homotopical universal property of cartesian products. Only the counted theorem and its author proof are retained; the author's parenthetical remark on the implicit product induction is background.

Allowed standard prerequisites (editorial).

- Cartesian products and the product induction principle allowing a function on a product of function types to be defined on pairs.
- The characterization of paths in product types and its uniqueness corollary (the retained context).
- Function extensionality: pointwise identified functions are identified as functions.
- The uniqueness principle for functions: a pair of functions applied pointwise is judgmentally the pair of the pointwise parts.

For instance, given types X, A, B, we have a function

$$(X \to A \times B) \to (X \to A) \times (X \to B) \tag{0.0.1}$$

defined by $f \mapsto (\mathsf{pr}_1 \circ f, \mathsf{pr}_2 \circ f)$.

For any elements $x, y : A \times B$ and a path $p : x =_{A\times B} y$, by functoriality we can extract paths $\mathsf{pr}_1(p) : \mathsf{pr}_1(x) =_A \mathsf{pr}_1(y)$ and $\mathsf{pr}_2(p) : \mathsf{pr}_2(x) =_B \mathsf{pr}_2(y)$. Thus, we have a function

$$(x =_{A\times B} y) \to (\mathsf{pr}_1(x) =_A \mathsf{pr}_1(y)) \times (\mathsf{pr}_2(x) =_B \mathsf{pr}_2(y)). \tag{0.0.2}$$

Theorem 0.0.3. *For any x and y, the function* (0.0.2) *is an equivalence.*

Theorem 0.0.4. (0.0.1) *is an equivalence.*

Proof. We define the quasi-inverse by sending (g, h) to $\lambda x.\, (g(x), h(x))$. (Technically, we have used the induction principle for the cartesian product $(X \to A) \times (X \to B)$, to reduce to the case of a pair. From now on we will often apply this principle without explicit mention.)

Now given $f : X \to A \times B$, the round-trip composite yields the function

$$\lambda x.\, (\mathsf{pr}_1(f(x)), \mathsf{pr}_2(f(x))). \tag{0.0.5}$$

By Theorem 0.0.3, for any $x : X$ we have $(\mathsf{pr}_1(f(x)), \mathsf{pr}_2(f(x))) = f(x)$. Thus, by function extensionality, the function (0.0.5) is equal to f.

On the other hand, given (g, h), the round-trip composite yields the pair $(\lambda x.\, g(x), \lambda x.\, h(x))$. By the uniqueness principle for functions, this is (judgmentally) equal to (g, h). □